Software

TYPES OF SOFTWARE

Definition of software

- Software is a set of instructions, technically referred to as programs, that perform operations and specific tasks based on the commands of the user.
- Every single task that a user intends to perform is regulated by software.
- Made of binary language (ones and zeroes), there is a variety of software for different tasks.

Types of software

- There is two types of software and these are:
  1. Application Software
  2. Operating system or System software

Application software

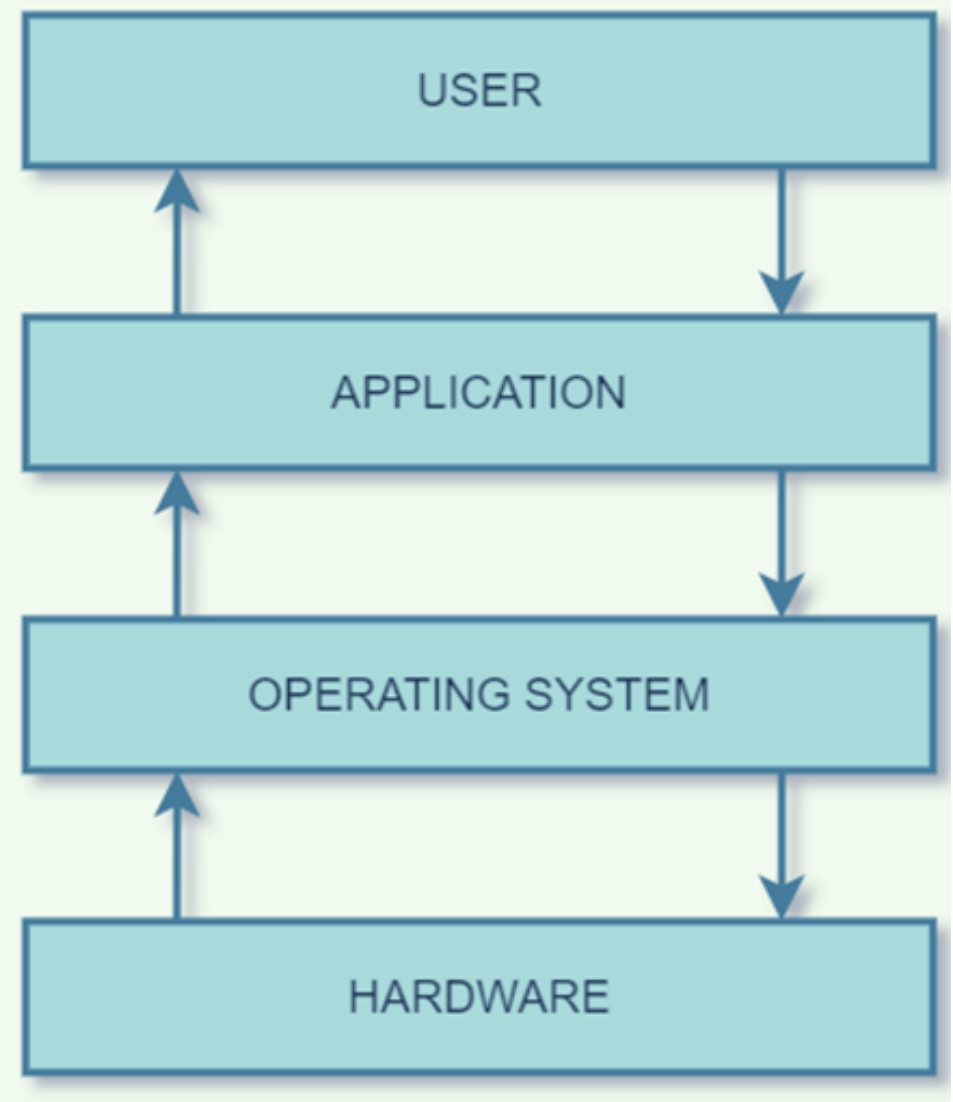
- Application software or application programs are end-user programs that serve specific functionality to help users accomplish certain tasks.
- This includes graphic designing, researching online, drafting documents, watching movies or playing games, and managing finance among others.
- Developers keep creating software applications based on the evolving needs of users.

Application software. Contd

- There are various types of application software, which are as follows-
- Word Processors: used for typing and documentation.
- Database software: used to perform various operations to quickly create, manage, organise, retrieve and modify data.
- Web Browsers: used for browsing the internet and their primary function is to allow users to fetch data across the web.

Operating System or System Software

- An operating system (OS) is software that runs on a computer.
- It manages all other applications and programs in a computer, and a boot program loads it into the computer.
- The OS enables applications to interact with a computer’s hardware.
- Application programs request services from the operating system through a designated application program interface (API).



Operating System or System Software. Contd

- The kernel is the software that contains the operating system's core components.
- Every computer must have at least one operating system installed to run other programs.

Why is an Operating System necessary for a computer?

- An operating system (OS) is essential for the functioning of any computer system.
- In the next slide are the key reasons why using an operating system is crucial:

1. Resource Management

- The OS manages the system's hardware and software resources, including the CPU, memory, storage, and peripheral devices.
- It ensures these resources are allocated efficiently and fairly among all running applications.

2. Interface

- Operating systems provide a user interface (UI), which can be graphical (GUI) or command-line (CLI).
- This interface allows users to interact with the computer system, execute programs, manage files, and perform various tasks efficiently.

3. Application Support

- The OS provides a stable and consistent environment for applications to run.
- It offers essential services and APIs (Application Programming Interfaces) that developers use to build and execute software applications.

4. File Management

- The OS manages files on storage devices, organizing data into directories and files.
- It provides mechanisms for reading, writing, and accessing files, ensuring data integrity and security.

5. Security and Access Control

- Operating systems implement security measures to protect data and system resources.
- They manage user accounts, permissions, and access controls to ensure that only authorized users can access certain resources and perform specific actions.

6. Process Management

- The OS executes multiple processes (programs) simultaneously.
- It schedules processes, allocates CPU time, and manages multitasking to ensure the system runs smoothly and efficiently.

7. Device Management

- The OS manages communication with peripheral devices like printers, scanners, and external drives.
- It provides device drivers and handles input/output operations, ensuring that devices function correctly with the system.

8. Networking

- Modern operating systems include networking capabilities that allow computers to connect to local networks and the internet.
- They manage network connections, data transfer, and communication protocols.

9. Error Handling

- The OS detects and responds to hardware and software errors.
- It provides error reporting and recovery mechanisms, ensuring system stability and reliability.

10. Support for Multitasking

- Operating systems enable multitasking, allowing multiple applications to run simultaneously. This increases productivity and allows users to perform various tasks concurrently.
- Using an operating system simplifies the complexity of managing computer hardware and software.
- It provides a user-friendly environment, essential services, and robust security, making it indispensable for modern computing.